adzymic

Take-Home Assignment: Build a Mini Creative Agent

AI / Agentic Engineer

Adzymic is a Singapore-based ad tech company. Our AgenX agent suite uses AI agents to create ads, sell publishers' ad space and buy media, built on open protocols like MCP and AdCP. This assignment is a small version of the work you'd do with us.

The task

Build a small multi-agent system that takes one size storyboard ad, a brand kit and a short brief, and produces the same ad in 5 new sizes. Every output must be checked automatically before it's marked as done and follow the brand guidelines etc..

Inputs

- **Master ad:** an story board ad with generic size
- **Brand kit:** a JSON file with colours, fonts and logo rules (open to add more stuff if needed)
- **Brief:** a short text prompt describing the campaign
- Open to have other inputs as well such as skills.md file etc.

The inputs include some realistic problems. How you notice and handle them is part of the assessment.

Output sizes

300×250	728×90	320×50	160×600	970×250

Requirements

1. **Multiple agents.** Use separate agents for different jobs, for example planning, building and QC. Explain why you split it that way.
2. **Automatic QC.** A QC agent runs on every output using a headless browser. At minimum, check:
   - No text overflow or cut-off text, and all elements inside the canvas
   - Logo size and spacing follow the brand kit
   - Only brand colours and fonts are used
   - File size is under 150 KB, and the click-through link works

   Failed outputs go back for a fix and are re-checked, with a limit on retries. Extra checks earn extra credit.
3. **Brand rules in code.** Every output is checked against the brand kit in code. Relying on prompts alone isn't enough.
4. **MCP server.** Expose the system with at least create_job, get_job_status and get_result. Jobs should run asynchronously.
5. **Trace log.** Record every step, model call and QC result for each job.

Bonus: compare your system with a simple single-prompt approach. Report the pass rate and cost per approved ad for both.

Rules

- Plan for about 3 hours of work. Submit within 5 days.
- Use any language, model or API. AI coding tools are allowed.(During review we will input our own API key with the latest model to check results.)
- Don't commit API keys to your repo.

What to submit

- A repo link with setup instructions that work in under 10 minutes
- All output ads, with screenshots
- A README of one page at most: your design, what didn’t work, and what you’d do with 2 more weeks
- A 5-minute screen recording of one job running from start to finish

What happens next

If your submission moves forward, we’ll schedule a 45-minute call. You’ll walk us through one job’s trace and make a small live change to your code.

Questions? Reply to the email this came with.